

Week 1 - Day 2a Supplementary: Variables, Printing, Commenting, Conditionals

Here are some more challenging concepts related to variables and conditionals. While not directly focused on printing or commenting, you should still try to use `print()` statements to help you code and write comments to clarify any confusing lines of code.

Some of these concepts may not have been learned in lecture, so feel free to ask for help if you get stuck or use tGood luck and have fun!

Exercise 1: Variables []

For the following exercises, try to answer **without running the code first**. Once you believe you have solved the questions, you may use `print()` to help check your answer.

1.1 What is `a`?

```
a = 3
b = 4
a = a * b
a
```

1.2 What is `c`?

```
a = 3
b = 6
c = b / a
a = b % c
b = a + b
c = b + c
c
```

1.3 What is `c`?

```
a = 2
b = 6
c = (a + b) // 3
a = c + a * b
b = (c + a) * b
c += a + b
```

1.4 What is length (number of characters) of `full_name`?

```
full_name = 'Jelani Nelson'
```

1.5 What is `full_name`?

```
first_name = 'Tamiru'
last_name = 'Alemnew'
full_name = first_name + last_name
```

1.6 Are `full_name` and `concatenated_name` equal?

```
full_name = 'Sonika Vuyyuru'
concatenated_name = 'Sonika' + 'Vuyyuru'
```

1.7 What is `b`?

```
a = 'hello'
b = a * 3
b
```

1.8 Can we compute `c` without errors? Why or why not?

```
a = 'hello'
b = 3
c = a + b
```

1.9 What is `a` and its type?

```
a = 'hello'
a = a > 'bye'
a += 1.0 - False
a *= a
```

1.10 The `input()` function (comes directly with Python) allows you to ask the user for input and store it as a variable. For example, if you run the following code, you will be prompted to enter your name, and then the variable `name` will be set to whatever you entered.

```
name = input()
```

Write code that asks the user for their name and then prints `"Hello, {name}!"`.

```
# TODO: Your code here
```

1.11 Like the previous question, but this time ask the user for their age and then print `"You are born in year [year]"`. For example, if the user enters 20, the output should be `"You are born in year 2004"`.

Watch out for the type of the input!

```
# TODO: Your code here
```

1.12 We can encode a point as a list of its two coordinates, for example, `p = [-3, 2]`.

Define two points `p0`, `p1`, and consider the rectangle where these points are the opposite points of a rectangle. We will use `p0_x` and `p0_y` as the x and y coordinates of the point; same for `p1_x` and `p1_y`. Compute the area of the rectangle.

Example:

- `p0_x, p0_y = -1, 1` and `p1_x, p1_y = 4, 11` should give $5 \times 10 = 50$.
- `p0_x, p0_y = 1, 1` and `p1_x, p1_y = 5, 4` should give $4 \times 3 = 12$.
- `p0_x, p0_y = 5, 4` and `p1_x, p1_y = 1, 1` should give $4 \times 3 = 12$.
- `p0_x, p0_y = 1, 4` and `p1_x, p1_y = 5, 1` should give $4 \times 3 = 12$.

```
# TODO: Your code here
```

```
p0_x, p0_y = -1, 1
p1_x, p1_y = 4, 11
```

Exercise 2: Conditionals []

Here we start with some conditional coding exercises before free response questions that test a deeper understanding of conditionals.

2.1 Define a **positive floating point** number `x`. Then, write code that computes and prints the **rounded** version of that `x`. Make sure it works for all values of `x`. **Do not use the built-in function `round` in Python**

For example:

- `x = 1.2` --> 1
- `x = 12.6` --> 13
- `x = 0.5` --> 1

Hint: You can **round down** a float to the nearest integer by converting it to an integer using the `int` function. Could that be useful here?

```
# TODO: Your code
```

2.2 Write some code that takes three numbers as inputs and prints the **largest** one.

```
# TODO: Your code
```

2.3 Write an if statement that checks if a year is a leap year.

```
# TODO: Your code
```

2.4 What will be the output of the following code snippet?

```
x = True
y = False
z = False

if not x or y:
    print("A")
elif not x or not y and z:
    print("B")
elif not x or y or not y and x:
    print("C")
else:
    print("D")
```

2.5 What will be the output of the following code?

```
x = 20
if x < 10:
    print("A")
elif x > 15:
    print("B")
if x % 2 == 0:
    print("C")
else:
    print("D")
```

2.6 What will be the output of the following code?

```
if 1 + 1 * 3 == 6:
    print("A")
elif 1 + 1 * 3 == 5:
    print("B")
else:
    print("C")
```

2.6 What will be the output of the following code?

```
x = 15
y = 20
if x > 10:
    if y > 15:
        print("A")
    else:
        print("B")
else:
    print("C")
```

2.7 Which of the following lines will never be printed, regardless of the value of `a`? (Could be more than one.)

```

a = int(input("Some number:"))
if a > 20:
    print("Big!")
elif a < 0:
    print("Negative.")
elif a > 30:
    print("Very Big.")
if a < 0:
    print("Still negative.")
elif a <= 0:
    print("Kinda negative.")
    if a == 5:
        print("Five.")
    if a == -5:
        print("Minus Five.")
    else:
        print("Not Minus Five.")
elif a >= 1:
    print("Nice number.")
else:
    print("Whatever.")

```

2.8 Find an input value so that the output for the two pieces of code below is different!

For which range of values is the output different?

```

# TODO: Your code here
length = ...

if length < 20:
    print("A bug!")
if length > 50:
    print("Snake is loooong...")
elif length >= 20:
    print("Could be a worm.")

if length < 20:
    print("A bug!")
elif length > 50:
    print("Snake is loooong...")
if length >= 20:
    print("Could be a worm.")

```

Exercise 3: Rewriting Conditionals []

3.1 Without using `elif` or `else`, write some code that, given the same input, prints the exact same output as the code given below.

Hint: The logical `and` might come in handy here.

```

size = float(input("Animal size (in cm):"))
if size < 1:
    print("That's an ant!")
elif size > 500:
    print("That's an elephant.")
elif size > 100:
    print("Is that a lion?")
else:
    print("That must be a hyena!")

```

That must be a hyena!

TODO: *Your code here*

3.2 Without nesting `ifs` (i.e. `ifs` within an `if` block), write some code that, given the same input, prints the exact same output as the code given below.

```

width = int(input("Width:"))
height = int(input("Height:"))
if width > 1:
    print("That's a good width!")
    if height > 1:
        print("Perfect size!")
    else:
        print("Could be a bit taller, but ok.")
else:
    print("That's too small!")

```

TODO: *Your code here*

Exercise 4: More Coding with Conditionals []

4.1 The Body Mass Index (BMI) is a person's weight in kilograms divided by the square of height in meters.

$$\text{BMI} = \text{Weight} / \text{Height}^2$$

Write a computer program that computes the BMI based on the above formula. Use the value of the `height` and `weight` variables given below to compute BMI and print the message accordingly

If the BMI value is: less than 18.5, print "You are underweight" between 18.5 and 24.9, print "You are in healthy range". greater than 25, print "You are overweight".

TODO: *Your code here*

4.2 Improve your code in the above question, to request for a user input (by using the built-in function `input()` we discussed above) and get user's height and weight to compute the BMI.

```
# TODO: Your code here
```

Exercise 5: Variable Casting []

5.1 What is `b` and what is its type?

```
a = '1'  
b = int(a) + 1
```

5.2 What is `b` and what is its type?

```
a = 1.33  
b = int(a)
```

5.3 What is `b` and what is its type?

```
a = 1.9  
b = int(a)
```

5.4 Get "hello3" in the variable `c` using `a` and `b`.

```
# TODO: Your code here  
a = 'hello'  
b = 3  
c = ...
```

5.5 What is `b`, `c`, `d`, `e`, `f`, and `g`? What are the types of `d`, `e`, `f`, and `g`? (You can later check the type of a variable using `type()`; but please work out the answers first.)

```
a = True  
b = int(a)  
c = float(a)  
d = a + b  
e = b + c  
f = a + c  
g = a + b + c
```

5.6 What is `a_as_bool`, `b_as_bool`, and `c_as_bool`?

```
a = 1  
b = 0  
c = -1  
a_as_bool = bool(a)  
b_as_bool = bool(b)  
c_as_bool = bool(c)
```

5.7 What is `a` and `b`?

```
a = bool('')
b = bool('Ken')
```

5.8 What is `a`?

```
a = bool('')
```

5.9 What is `a` and `b`?

```
a = str(False)
b = str(bool(-2))
```

5.10 What is `a`, `b`, `c`, `d`? What is the `type` of them?

```
a = True + 1
b = False * 10
c = True + True - False
d = True * 1.0 - 1
```

5.11 What is `a` and `b`? What is the `type` of them?

```
a = True * 10 * 'd'
b = False * 5 * 'e'
```

5.12 What is `a` and `b`?

```
a = bool(3.14)
b = bool(-0.000000000001)
```

Exercise 6: Python Printing []

Let's go over some other ways to print with Python! To start, assign your name to the `name` variable. Run the following code blocks.

```
# TODO: Your code here
name = ""
```

You've seen string concatenation before!

```
print("Hello " + name)
```

You can add many parameters to the `print` function, and they will be separated by the separator, which is a space by default.

```
print("You did great work today", name)
print("Congrats ", name, " on finishing!", sep="")
```

Python also has a `format()` method. `print(f"...")` works similarly but you can insert the variables you want the values of straight into the function. This may be easier to format if you want to print text with many variables.

```
print("Hey {0}, there were {1} long exercises in this  
notebook!".format(name, 6))  
print(f"Hey {name}, there were {6} long exercises in this notebook!")  
print(f"See you next time, {name}!")
```